

Computer Programming

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Course Summary

"The value of a college education is not the learning of many facts but the training of the mind to think." Albert Einstein

The first course for software engineering majors and anyone seeking a rigorous introduction. Develops computational thinking and problem-solving skills by programming, and exposes students to variety of other topics from computer science and its applications.

Prerequisites: High school mathematics and Excitement

Course Topics

Chapter 1: Introduction

- Introduction to computations
- Introduction to programs

Chapter 2: Variables, Expressions and Statements

- Primitive data types and values
- Variables
- Data Types & Casting
- Booleans and Expressions

Chapter 3: Functions

- What are functions?
- Built-in Functions
- Function calls
- Parameters and Arguments
- Flow of execution
- Function composition

Chapter 4: Conditionals

- Boolean Expression and Logical Operators
- Conditional Execution
- Chained and Nested Conditions
- Keyboard input and Defensive Programming

Chapter 5: Functions (Not a typo ... functions again)

- User-Defined Functions
- Fruitful and void functions

- Return vs Print
- Function Scope
- Decomposition and Abstractions

Chapter 6: Iteration

- Looping Basics
- The while loop
- Break and Continue
- For Loops, Nested Loops

Chapter 7: Strings

- String as a sequence
- Indexing
- Mutability and Immutability
- Searching
- String Comparison

Chapter 8: Data Structures

- What are Data Structures
- Lists
- Numerical Indexing
- Dictionaries
- Tuples and Sets

Chapter 9: Persistence

- File
- Reading and Writing
- Common file formats
- Pickling

Chapter 10: Object Oriented Programming

- Class and Object
- Object Modelling

Assessment

Mid exam: 30%

Assignment: 20%

Final Exam: 50%

Text Book:

Allen Downey. How to Think Like a Computer Scientist (2

nd edition).

Reference:

Liang. Introduction to Python Programming

Online Course

Python For Everybody Specialization (Course 1 and 2). URL
(<https://www.coursera.org/specializations/python#courses>)